

Curriculum Vitae

Jinho Byun

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RESEARCH EXPERIENCE

Mar. 2025 – Present	Research Professor, Dept. of Energy Engineering, KENTECH
Mar. 2023 – Mar. 2025	Postdoctoral Researcher, Dept. of Energy Engineering, KENTECH
Aug. 2022 – Jan. 2023	R&D Engineer, Wonik IPS
Mar. 2022 – Jul. 2022	Postdoctoral Researcher, Dept. of Physics, Pusan National University (Advisor: Prof. Jaekwang Lee)

EDUCATION

Sep. 2017 – Feb. 2022	Ph.D., Physics, Pusan National University. Advisor: Prof. Jaekwang Lee. Thesis: “Emergent Phenomena at Polar Oxides and Their Applications.”
2012 – 2017	B.A., Physics, Pusan National University

HONORS AND AWARDS

2021	Outstanding Poster Award, Korean Physical Society (KPS) — “Interface engineered TER optimization in HfO ₂ -based FTJ”
2021	Excellent Academic Award, Pusan National University
2020	Humantech Paper Award, Samsung Electronics — “Anisotropic Evaporation of ZnO Observed by In-situ Cs-Corrected High Resolution TEM”
2019	Outstanding Poster Award, Korea Institute for Advanced Study (KIAS) — “First-principles study of the HfO ₂ -based ferroelectric tunnel junction”
2017	Outstanding Presentation Award, Korean Physical Society (KPS) — “Design of high-coercivity Fe ₁₆ -xAl _x N ₂ alloy”
2017 – 2020	Combined-course Scholarship (full tuition), Pusan National University
2012 – 2017	National Excellence Scholarship (Natural Sciences and Engineering), Korean Government (full tuition)

RESEARCH INTEREST

Methods: deep learning for microscopy, ab initio multislice image simulation, ab initio / machine-learning-force-field molecular dynamics, phonon properties, DFT

Materials systems: oxide surfaces and interfaces, perovskites, 2D materials

PUBLICATIONS

¶ equal/co-first contribution; * corresponding author.

Recent publications and accepted papers

1. Y. Xing¶, **J. Byun¶**, K. T. Kang, J. Hwang, J. Lee, W. S. Choi, S. H. Oh*. “Atomic-Scale Visualization of Vacancy-Ordered Intermediates and Pathway Selection in the Perovskite–Infinite-Layer Transformation.”

Journal of the American Chemical Society **148**(34), 36770–36781 (2026). doi:10.1021/jacs.6c08044 **Published 2 September 2026.**

2. Hoin Lee[¶], Joonbong Lee[¶], **J. Byun[¶]**, Sang-Hyeok Yang[¶], Ayoung Cho, Sangwoo Lee, Dae Haa Ryu, Hyunbin Chung, Hee Seo Yun, Moon Seop Choi, Yesul Choi, Sungkyun Park, Min-Hyung Jung, Dongwon Lee, Yoon-Uk Heo, Hu Young Jeong, Hongseok Kim, Shinhyeong Lee, Yunseok Kim, Y.-M. Kim*, J. Lee*, T. Choi*. “Hybrid Ferroelectric Tunnel Junctions with Intrinsic Nonlinearity and Self-Rectification via Interfacial Reconstruction.” *Advanced Functional Materials* **e78527** (2026). doi:10.1002/adfm.78527 **Published 16 September 2026.**
3. J. Jeong[¶], **J. Byun[¶]**, X. Xu, A. Gruverman, J. Lee, S. H. Oh*. “Direct Visualization of Interfacial Defect Effects on Polarization Switching in BaTiO₃ Tunnel Junctions.” *Science Advances* **12**(38) (2026). doi:10.1126/sciadv.aeh5401 **Published September 2026.**

Under review

4. K. Lee[¶], Y. Xing[¶], **J. Byun[¶]**, M. Jeong[¶], J. Seo, J. Bang, E. Lee*, S. H. Oh*. “Visualizing Oxygen Vacancy-Induced Dielectric Dead Layer at Metal-Oxide Interfaces by In-situ Multimodal Electron Microscopy.” *Science Advances* (under review, major revision).
5. Y. Xing[¶], **J. Byun[¶]**, J. Seo, Z. Wang, Y. Jin, S. Kim, J. Lee, S. H. Oh*. “Geometry-driven surface electron dichotomy via ordered subsurface defects.” *Nature Physics* (under review).

Peer-reviewed journal articles (published)

6. Y. Xing, I. Kim, K. T. Kang, **J. Byun**, W. S. Choi, J. Lee, S. H. Oh*. “Monitoring the formation of infinite-layer transition metal oxides through in situ atomic-resolution electron microscopy.” *Nature Chemistry* **17**, 66–73 (2025). doi:10.1038/s41557-024-01617-7 [Author Correction: doi:10.1038/s41557-025-01816-w]
7. J. Seo, H. Lee, K. Eom, **J. Byun**, T. Min, J. Lee, K. Lee, C.-B. Eom, S. H. Oh. “Field-induced modulation of two-dimensional electron gas at LaAlO₃/SrTiO₃ interface by polar distortion of LaAlO₃.” *Nature Communications* **15**, 5268 (2024). doi:10.1038/s41467-024-48946-2
8. J. Hwang, **J. Byun**, H. Kim, J. Lee. “High-mobility two-dimensional electron gas at the PbZr_{0.5}Ti_{0.5}O₃/BaSnO₃ heterostructure.” *J. Korean Phys. Soc.* **84**(1), 67–72 (2024). doi:10.1007/s40042-023-00978-5
9. Z. Wang[¶], **J. Byun[¶]**, Z. Wang, Y. Xing, J. Seo, J. Lee, S. H. Oh*. “Direct Observation of Atomic Step-Assisted Stabilization of Polar Surfaces.” *Advanced Materials* **35**(40), 2303051 (2023). doi:10.1002/adma.202303051
10. M. H. Sheen, Y. H. Ko, D. U. Kim, N. R. Ahn, C. H. Lee, D. H. Kim, R. Kim, I. P. Kim, et al., **J. Byun**. “Light emitting element, method for fabricating the same and display device.” US Patent App. 18/093,959 (2023). [Patent — no DOI]
11. Z. Wang[¶], **J. Byun[¶]**, S. Lee, J. Seo, B. Park, J. Kim, H. Y. Jeong, J. Bang, J. Lee, S. H. Oh*. “Vacancy driven surface disorder catalyzes anisotropic evaporation of polar ZnO (0001) surface.” *Nature Communications* **13**, 5616 (2022). doi:10.1038/s41467-022-33353-2
12. Y. Ha, **J. Byun**, J. Lee, J. Son, Y. Kim, S. Lee. “Infrared Transparent and Electromagnetic Shielding Correlated Metals via Lattice-Orbital-Charge Coupling.” *Nano Letters* **22**, 6573–6579 (2022). doi:10.1021/acs.nanolett.2c01487
13. M. Sheen, Y. Ko, D. Kim, J. Kim, **J. Byun**, K. Y. Yeon, D. Kim, J. Jung, J. Choi, R. Kim, J. Yoo, I. Kim, C. Joo, N. Hong, J. Lee, S. H. Jeon, S. H. Oh, J. Lee, N. Ahn*, C. Lee*. “Highly efficient blue InGaN nanoscale light-emitting diodes.” *Nature* **608**, 56–61 (2022). doi:10.1038/s41586-022-04933-5
14. K. Kang, **J. Byun**, M. Jeon, G. Jo, Y. K. Baek, J. Lee, H. Jeon. “Effect of sintering time on electronic properties of strontium hexaferrite.” *Ceramics International* **48**(9), 12476–12482 (2022). doi:10.1016/j.ceramint.2022.01.113
15. K. Lee, **J. Byun**, K. Park, S. Kang, M.-S. Song, J. Park, J. Lee, S. Chae*. “Giant Tunneling Electroresistance in Epitaxial Ferroelectric Ultrathin Films Directly Integrated on Si.” *Applied Materials Today* **26**, 101308 (2022). doi:10.1016/j.apmt.2021.101308

16. J. Lee, M. S. Song, W.-S. Jang, **J. Byun**, H. Lee, M. H. Park, J. Lee, Y.-M. Kim*, S. C. Chae*, T. Choi*. "Modulating the ferroelectricity of hafnium zirconium oxide ultrathin films via interface engineering to control the oxygen vacancy distribution." *Advanced Materials Interfaces* **9**(7), 2101647 (2022).
[doi:10.1002/admi.202101647](https://doi.org/10.1002/admi.202101647)
17. Y. Ha, **J. Byun**, J. Lee, S. Lee. "Design Principles for the Enhanced Transparency Range of Correlated Transparent Conductors." *Laser & Photonics Reviews* **15**(5), 2000444 (2021). [doi:10.1002/lpor.202000444](https://doi.org/10.1002/lpor.202000444)
18. Y.-I. Kim¶, M. Jeong¶, **J. Byun**¶, S.-H. Yang, W. Choi, W.-S. Jang, J. Jang, K. Lee, Y. Kim, J. Lee, E. Lee, Y.-M. Kim*. "Atomic-scale identification of invisible cation vacancies at an oxide homointerface." *Materials Today Physics* **16**, 100302 (2021). [doi:10.1016/j.mtphys.2020.100302](https://doi.org/10.1016/j.mtphys.2020.100302)
19. J. Jo¶, **J. Byun**¶, J. Lee, D. Choe, I. Oh, J. Park, M.-J. Jin, J. Lee*, J.-W. Yoo*. "Emergence of Multispinterface and Antiferromagnetic Molecular Exchange Bias via Molecular Stacking on a Ferromagnetic Film." *Advanced Functional Materials* **30**(11), 1908499 (2020). [doi:10.1002/adfm.201908499](https://doi.org/10.1002/adfm.201908499)
20. J. Jo, **J. Byun**, I. Oh, J. Park, M.-J. Jin, B.-C. Min, J. Lee*, J.-W. Yoo*. "Molecular tunability of magnetic exchange bias and asymmetrical magnetotransport in metalloporphyrin/Co hybrid bilayers." *ACS Nano* **13**(1), 894–903 (2019). [doi:10.1021/acs.nano.8b08689](https://doi.org/10.1021/acs.nano.8b08689)
21. S.-J. Kwon, B.-M. Jung, T. Kim, **J. Byun**, J. Lee, S.-B. Lee, U.-H. Choi*. "Influence of Al₂O₃ Nanowires on Ion Transport in Nanocomposite Solid Polymer Electrolytes." *Macromolecules* **51**(24), 10194–10201 (2018).
[doi:10.1021/acs.macromol.8b01603](https://doi.org/10.1021/acs.macromol.8b01603)
22. T. T. Ly, G. Duvjir, T. Min, **J. Byun**, T. Kim, M. M. Saad, N. T. M. Hai, S. Cho, J. Lee*, J. Kim*. "Atomistic study of the alloying behavior of crystalline SnSe_{1-x}S_x." *Physical Chemistry Chemical Physics* **19**(32), 21648–21654 (2017). [doi:10.1039/c7cp03481d](https://doi.org/10.1039/c7cp03481d)

Preprint

- **J. Byun**¶, K. Lee¶, M. Jeong, E. Lee, J. Bang, H. Kim, G. H. Gu*, S. H. Oh*. "Imaging 3D polarization dynamics via deep learning 4D-STEM." *arXiv:2506.06598* (2025). [doi:10.48550/arXiv.2506.06598](https://doi.org/10.48550/arXiv.2506.06598)

CONFERENCE PRESENTATIONS

Talks

1. J. Byun, K. Lee, J. Oh, D. Park, H. Yoo, G. H. Gu, S. H. Oh. (2026). "Unveiling 3D Atomic Structure in Twisted Heterobilayers via Deep Learning-Assisted Ptychography." 21st International Microscopy Congress (IMC21), Liverpool, UK.
2. J. Byun, K. Lee, J. Bang, H. Kim, D. Park, H. Yoo, G. H. Gu, S. H. Oh. (2026). "Beyond the Image: Transitioning from Conventional TEM to 4D-STEM for 3D Analysis." Invited seminar, Korea Institute of Energy Research (KIER).
3. J. Byun, K. Lee, J. Bang, J. Lee, J. Lee, H. Kim, G. H. Gu, S. H. Oh. (2026). "Visualizing Polarization Switching and Electrical Dead Layers via In-situ Multimodal STEM Biasing." 2026 Korean Dielectrics/Ferroelectrics Society Winter Conference & Korea–China Ferroelectrics Symposium, Muju, Korea.
4. J. Byun, K. Lee, J. Bang, J. Lee, J. Lee, H. Kim, G. H. Gu, S. H. Oh. (2026). "Deep Learning Analysis of 4D-STEM Reveals 3D Polarization and Switching in Ferroelectric Capacitors." 33rd Korean Conference on Semiconductors (KCS 2026), High1 Resort, Wonju, Korea.
5. J. Byun, K. Lee, J. Bang, H. Kim, D. Park, H. Yoo, G. H. Gu, S. H. Oh. (2025). "Unveiling 3D Structural Information via Deep Learning of 4D-STEM." 2025 Korean Society of Microscopy (KSM) Fall Conference, Cheongju, Korea.
6. J. Byun, Y. Xing, S. H. Oh. (2025). "Emergence of 2DEG from Geometric Constraints in the SrTiO₃ Subsurface." 2025 Korean Physical Society Fall Meeting, Kimdaejung Convention Center, Gwangju, Korea.

7. J. Byun, K. Lee, D. Park, H. Yoo, G. H. Gu, S. H. Oh*. (2025). "Machine Learning-Driven 3D Sectioning and Analysis in Electron Microscopy." 2025 Korean Physical Society Spring Meeting, Daejeon Convention Center, Daejeon, Korea.
8. J. Byun, K. Lee, D. Park, H. Yoo, G. H. Gu, S. H. Oh. (2024). "Machine Learning-Driven 3D Sectioning and Analysis in Electron Microscopy." 2024 MRS Fall Meeting, Boston, MA, USA.
9. J. Byun, K. Lee, J. Bang, G. H. Gu, S. H. Oh. (2024). "Three-Dimensional Polarization Mapping with Pico-Meter Precision via Deep Learning of 4D-STEM." 2024 Korean Physical Society Fall Meeting, Yeosu EXCO, Yeosu, Korea.
10. J. Byun, K. Lee, D. Park, H. Yoo, G. H. Gu, S. H. Oh. (2024). "Unveiling 3D Structural Information via Deep Learning of 4D-STEM." NanoKorea 2024, KINTEX, Goyang, Korea.
11. J. Byun, K. Lee, D. Park, H. Yoo, G. H. Gu, S. H. Oh*. (2024). "Novel 3D Atomic Imaging Method in Electron Microscopy: Integrating Machine Learning with 4D-STEM." 21st International Symposium on the Physics of Semiconductors and Applications (ISPSA 2024), Jeju, Korea.
12. J. Byun, K. Lee, G. H. Gu, S. H. Oh. (2023). "Interpreting In-situ 4D-STEM Data for Accurate Measurement of Ferroelectric Polarization using Machine Learning Technique." 2023 MRS Fall Meeting, Boston, MA, USA.
13. J. Byun, K. Lee, G. H. Gu, S. H. Oh. (2023). "Interpreting In-situ 4D-STEM Data for Accurate Measurement of Ferroelectric Polarization using Machine Learning Technique." 20th International Microscopy Congress (IMC20), BEXCO, Busan, Korea.
14. J. Byun, J. Lee. (2022). "Enhanced magnetic anisotropy in $\text{Fe}_{16-x}\text{Al}_x\text{N}_2$ alloy." Talk presented at the Korean Magnetism Society Conference.
15. J. Byun, Z. Wang, S. Oh, J. Lee. (2021). "Novel Stabilization Mechanism on Polar Oxide Surface." 2021 Materials Research Society Fall Meeting.
16. J. Byun, Z. Wang, S. Oh, J. Lee. (2021). "Novel Stabilization Mechanism on Polar Oxide Surface." 2021 Korean Physics Society Fall Meeting.
17. J. Byun, J. Lee. (2021). "Giant Nonlinear Tunneling Current in HfO_2 -based Anti-Ferroelectric Tunnel Junction." Electronic Materials and Applications Conference 2021.
18. J. Byun, J. Lee, T. Choi, J. Lee. (2021). "Interface engineered TER optimization in HfO_2 -based FTJ." 2021 Korean Physics Society Spring Meeting.
19. J. Byun, J. Lee, T. Choi, J. Lee. (2020). "DFT investigation on structural stability on the ferroelectric HfO_2 on the electrode." 2020 Korean Physics Society Fall Meeting.
20. J. Byun, T. Min, J. Lee. (2020). "Nonlinear current-voltage behavior in HfO_2 -based ferroelectric domain-wall tunnel junction." 2020 Korean Ceramic Society Spring Meeting.
21. J. Byun, T. Min, J. Lee. (2020). "Giant Nonlinear Tunneling Current in HfO_2 -based Anti-Ferroelectric Tunnel Junction." American Physical Society March Meeting 2020.
22. J. Byun, T. Min, J. Lee. (2019). "First-principles studies of the effects of oxygen vacancies on the HfO_2 -based ferroelectric tunnel junction." 2019 Materials Research Society Fall Meeting.
23. J. Byun, T. Min, J. Lee. (2019). "First-principles studies of the HfO_2 -based ferroelectric tunnel junction." 5th International Conference on Advanced Electromaterials.
24. J. Byun, T. Min, J. Lee. (2019). "Design and modeling of optimal HfO_2 -based ferroelectric tunneling junction." 2019 Korean Physics Society Fall Meeting.
25. J. Byun, T. Min, J. Lee. (2019). "Design and modeling of optimal HfO_2 -based ferroelectric tunneling junction." Asia-Pacific PFM 2019 Conference.

26. J. Byun, T. Min, J. Lee. (2019). "Complex bandstructure calculation for the ferroelectric tunneling junction current." 2019 Korean Physics Society Spring Meeting.
27. J. Byun, T. Min, J. Lee. (2018). "First-principles studies of the effects of effective mass on the HfO₂-based ferroelectric tunnel junction." 2018 Korean Physics Society Fall Meeting.
28. J. Byun, T. Min, J. Lee. (2018). "Novel high coercivity Fe_{16-x}Al_xN₂ alloy." IcAUMS 2018.
29. J. Byun, T. Min, J. Lee. (2018). "High coercivity Fe_{16-x}Al_xN₂ alloy design." Korean Magnetics Society 2018 Spring Conference.
30. J. Byun, T. Min, J. Lee. (2018). "Novel high coercivity Fe_{16-x}Al_xN₂ alloy." American Physical Society March Meeting 2018.
31. J. Byun, T. Min, J. Lee. (2017). "First-principles modeling of electron tunneling through ferroelectric tunnel junction." 2017 KPS Fall Meeting.
32. J. Byun, T. Min, J. Lee. (2017). "Design of Fe_{16-x}Al_xN₂ with high-coercivity." 2017 KPS Spring Meeting.

Posters

1. J. Byun, K. Lee, J. Bang, H. Kim, G. H. Gu, S. H. Oh. (2025). "Imaging 3D Polarization Dynamics via Deep Learning 4D-STEM." 2025 Korean Society of Microscopy (KSM) Spring Meeting, Yonsei University, Seoul, Korea.
2. J. Byun, K. Lee, G. H. Gu, S. H. Oh. (2023). "Interpreting In-situ 4D-STEM Data for Accurate Measurement of Ferroelectric Polarization using Machine Learning Technique." Internal KENTECH research symposium.
3. J. Byun, K. Lee, G. H. Gu, S. H. Oh. (2023). "Interpreting In-situ 4D-STEM Data for Accurate Measurement of Ferroelectric Polarization using Machine Learning Technique." Samsung Global Technology Symposium (GTS) 2023, Samsung Electronics R&D Campus, Korea.
4. J. Byun, J. Jo, I. Oh, J. Park, M.-J. Jin, B.-C. Min, J.-W. Yoo, J. Lee. (2019). "First principles study of magnetic exchange interaction in Metalloporphyrin/Co interface." 3rd Workshop on Functional Materials Science.
5. J. Byun, T. Min, J. Lee. (2019). "First-principles studies of the HfO₂-based ferroelectric tunnel junction." 15th KIAS Electronic Structure Calculation Workshop.
6. J. Byun, T. Min, J. Lee. (2019). "Ab initio study of the role of effective mass in ferroelectric tunneling junction." 15th Dielectric Union Symposium.
7. J. Byun, T. Min, J. Lee. (2019). "First-principles studies of the effects of oxygen vacancy and effective mass on the HfO₂-based ferroelectric tunnel junction." 10th APCTP Workshop on Multiferroics.
8. J. Byun, T. Min, J. Lee. (2018). "First-principles studies of the effects of oxygen vacancies on the HfO₂-based ferroelectric tunnel junction." 12th Japan-Korea Conference on Ferroelectrics.

OTHER

- Military service (completed)
- 2021 — President of the (Physics Dept.) Student Council
- 2018 — KIAS CAC Summer School on Scientific Computing and Machine Learning
- 2017 — Python study manager for graduate students